

Operator ETL

Specification · Architecture · Build · Evaluation

Version 1.0 · August 2026

Local-first medallion pipeline: intake → warehouse → insights

Repository: <https://github.com/khaosans/operator-etl>

Problem

- Silent bad data — invalid rows appear in dashboards without visibility
- No idempotency — re-running the same file double-counts records
- Tight coupling — each new source needs a new script
- Overconfident outputs — charts render when upstream data is broken
- No audit trail — operators can't answer what ran, when, or what failed

Goals & non-goals

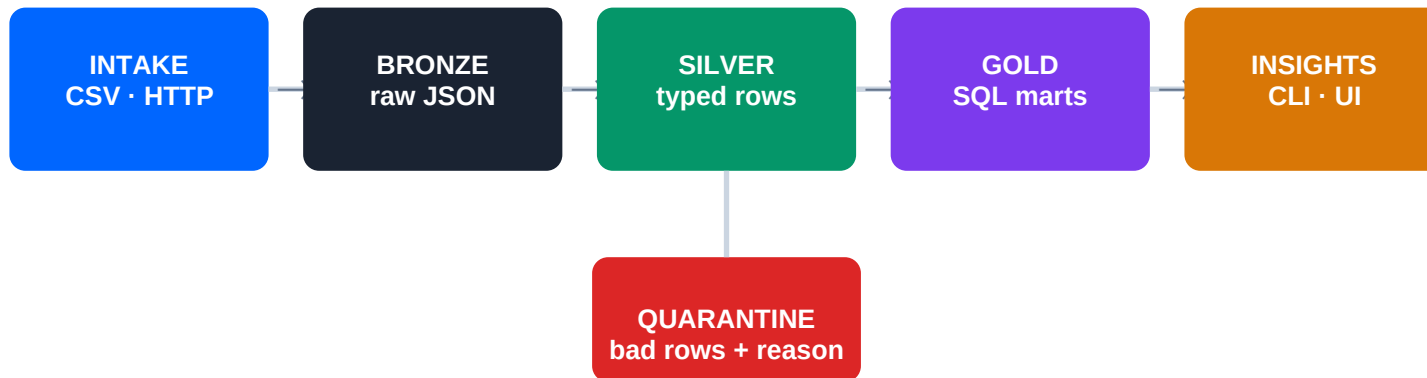
Goals (v1)

- Trustworthy intake — raw preserved, validate before silver
- Operator clarity — run logs, quarantine reasons
- Extensible sources via YAML registry
- Local-first — DuckDB on disk, no cloud required
- Deterministic ETL — Python + SQL, no LLM in critical path

Non-goals (v1)

- Cloud warehouse / hosted dashboard
- Airflow, Spark, Kafka, streaming
- LLM-generated insights
- Multi-tenant SaaS
- Domain adapters (Substack, finance) — later via registry

- Drop a CSV or call an API → store raw data immutably in bronze
- Validate and type rows into silver; reject bad rows to quarantine
- Compute gold metrics in SQL; surface KPIs in CLI or dashboard
- Withhold insights when quality gates fail — fail closed, not optimistic
- New sources = one registry entry; pipeline runner stays the same



Quality gate: KPIs withheld if quarantine rate > 35% · freshness > 7 days · silver empty

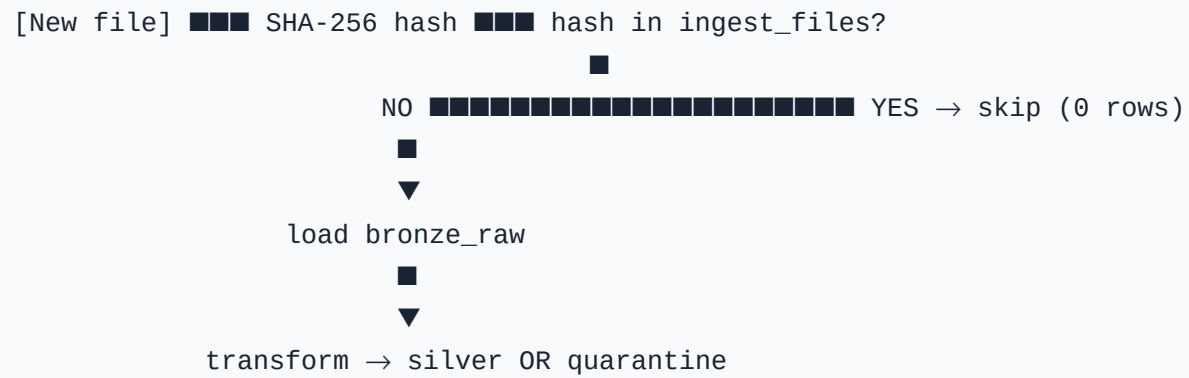
Layer	Responsibility	Technology
Extract	Read files/APIs, SHA-256 hash	Python, httpx
Bronze	Immutable raw JSON + metadata	DuckDB
Transform	Validate, type, dedupe	Pydantic v2
Quarantine	Isolate bad rows + error reason	DuckDB
Gold	Aggregated SQL marts	DuckDB SQL
Insights	KPIs when quality passes	CLI, Streamlit
Orchestration	Run logging, CLI commands	Typer

Source	Kind	Input	Purpose
demo	csv	samples/orders.csv	Demo & tests
inbox	csv_dir	drops/inbox/*.csv	Operator drop folder
http	http	GET JSON list	API intake stub

Table / mart	Contents
ingest_files	File hash registry (idempotency)
pipeline_runs	Run audit: counts, status, timestamps
bronze_raw	Raw JSON + _source, _ingested_at, _row_num
silver_orders	Validated typed business rows
quarantine_orders	Rejected rows + error message
gold_kpis / gold_volume_daily / gold_top_skus	Aggregated metrics
gold_quality	Quarantine rate, freshness, row counts

Field	Type	Rule
order_id	string	Required, non-empty, unique
customer_id	string	Required
ordered_at	datetime	Parseable ISO timestamp
amount	float	Must be > 0
sku	string	Required
status	string	Required

Check	Default	On failure
Quarantine rate	$\leq 35\%$	KPIs withheld
Freshness	≤ 7 days since ingest	KPIs withheld
Silver rows	> 0	KPIs withheld



```
$ uv run etl run --source demo
run ok source=demo rows_in=21 silver=17 quarantined=4
```

Operator ETL insights

```
Quality: PASS quarantine_rate=19.1%
```

KPIs

orders	17
customers	10
revenue	1373.82
avg order	80.81

Top SKUs

SKU-PRO	orders=5	revenue=967.40
SKU-WIDGET	orders=6	revenue=225.82

Orders

17

Customers

10

Revenue

1373.82

Gate

PASS

Quarantine (4 rows): empty order_id · invalid date · negative amount · non-numeric amount

Volume over time



Time	Volume
1	10
2	15
3	12
4	20
5	18
6	25

PRODUCTS

PRO

WIDGET

GADGET

- 1 Scaffold**
Repo · uv · DuckDB · CLI skeleton · source registry YAML
- 2 Data plane**
CSV/HTTP extract · bronze load · SHA-256 idempotency · run log
- 3 Transform**
Pydantic contract · silver load · quarantine table · dedupe
- 4 Gold + insights**
SQL marts · quality gate · CLI insight command
- 5 Dashboard**
Streamlit KPI cards · charts · quarantine expander · run history
- 6 Evaluate**
pytest: idempotency · quarantine · HTTP · quality gate block

```
operator-etl/  
  pipelines/demo.yaml    ← source registry  
  samples/               ← demo CSV + HTTP JSON  
  drops/inbox/           ← operator CSV drop  
  sql/marts/             ← gold SQL  
  src/operator_etl/      ← extract · load · transform · insights  
  dashboard/app.py       ← Streamlit UI  
  tests/                 ← pytest suite  
  warehouse/operator.duckdb (gitignored)
```

#	Criterion	Status
1	CSV in drop folder runs end-to-end	Met
2	Invalid rows → quarantine	Met
3	KPIs: freshness, volume, SKU breakdown	Met
4	Re-ingest same file → 0 duplicate rows	Met
5	Quality gate blocks bad KPIs	Met
6	HTTP JSON source works	Met
7	Automated test coverage	9/9 pass

Evaluation — test matrix

Test	Validates	Result
test_ingest_is_idempotent	Hash skip on re-run	PASS
test_quarantine_invalid_rows	4 bad rows isolated	PASS
test_run_builds_gold	KPIs + gate pass	PASS
test_quality_gate_blocks	KPIs blocked > threshold	PASS
test_http_*	JSON extract + warehouse	PASS
test_source_registry	demo, inbox, http	PASS

Metric	Value
Input rows (demo CSV)	21
Silver (valid)	17
Quarantined	4
Quarantine rate	19.1% → gate PASS
Re-run same file	0 new rows (hash skip)
Revenue (gold)	1373.82
Top SKU	SKU-PRO

CONTROL PLANE (v2)

LangGraph · checkpoints · HITL · critic · tool allowlists

POLICY PLANE (v2)

PII scan · token vault · redacted views · spend budgets

DATA PLANE (v1)

Extract · bronze · silver · quarantine · gold SQL

Phase	Capability
v1.0	Medallion ETL + CLI + dashboard (current)
v1.1	Real source adapters (Substack, finance CSV)
v1.2	Postgres/Supabase + scheduled runs
v2.0	LangGraph · PII gate · insight agent · critic · HITL
v2.1	Hosted dashboard · Langfuse observability

- Adopt v1 as the deterministic data plane — boring, testable, trustworthy
- Confirm canonical schema for first production data source
- Approve quality gate thresholds before live use
- Add v1.1 source adapter next; defer agentic layer until metrics are stable
- v2 agents orchestrate on top — they never replace ETL or bypass quarantine

Operator ETL v1
Intake → warehouse → insights
Questions?